

Medical Knowledge Base

Comprehensive Reference for AI-Assisted Medical Q&A;

1. Diabetes Mellitus

Definition

Diabetes is a chronic disease that affects how the body processes blood sugar (glucose). It occurs when the body either does not produce enough insulin or cannot effectively use it, resulting in high levels of glucose in the blood.

Types

- **Type 1 Diabetes:** An autoimmune condition where the immune system attacks insulin-producing beta cells in the pancreas. Patients require lifelong insulin therapy.
- **Type 2 Diabetes:** The most common form, characterized by insulin resistance. Often linked to obesity, physical inactivity, and genetics.
- **Gestational Diabetes:** Develops during pregnancy and usually resolves after delivery, but increases the risk of Type 2 diabetes later.
- **Prediabetes:** Blood sugar levels are higher than normal but not high enough to be classified as Type 2 diabetes.

Symptoms

- Frequent urination (polyuria)
- Increased thirst (polydipsia)
- Unexplained weight loss
- Extreme fatigue
- Blurred vision
- Slow-healing sores
- Frequent infections

Diagnosis

Test	Normal	Prediabetes	Diabetes
Fasting Blood Sugar	< 100 mg/dL	100–125 mg/dL	$\geq$ 126 mg/dL
HbA1c	< 5.7%	5.7–6.4%	$\geq$ 6.5%
2-hr Glucose (OGTT)	< 140 mg/dL	140–199 mg/dL	$\geq$ 200 mg/dL

Treatment

- Lifestyle modifications: healthy diet, regular exercise, weight management
- Oral medications: Metformin, SGLT-2 inhibitors, GLP-1 receptor agonists
- Insulin therapy (essential for Type 1, sometimes needed for Type 2)
- Blood sugar monitoring and regular HbA1c checks
- Management of complications: cardiovascular, renal, retinal

2. Hypertension (High Blood Pressure)

Definition

Hypertension, also known as high blood pressure, is a condition in which the force of blood against artery walls is consistently too high. It is defined as a sustained blood pressure of 130/80 mmHg or higher. It significantly increases the risk of heart disease, stroke, and kidney failure.

Classification

Category	Systolic (mmHg)	Diastolic (mmHg)
Normal	< 120	< 80
Elevated	120–129	< 80
Stage 1 Hypertension	130–139	80–89
Stage 2 Hypertension	≥ 140	≥ 90
Hypertensive Crisis	> 180	> 120

Risk Factors

- Age (risk increases after 65)
- Family history
- Obesity or overweight
- High sodium diet
- Physical inactivity
- Tobacco use and excessive alcohol
- Chronic conditions: diabetes, kidney disease

Treatment

- Lifestyle: DASH diet, reduce sodium, exercise 30 min/day, limit alcohol
- ACE inhibitors or ARBs (e.g., lisinopril, losartan)
- Calcium channel blockers (e.g., amlodipine)
- Diuretics (e.g., hydrochlorothiazide)
- Beta-blockers when indicated

3. Asthma

Definition

Asthma is a chronic respiratory condition where the airways become inflamed and narrow, making breathing difficult. It is characterized by recurring episodes of wheezing, breathlessness, chest tightness, and coughing, particularly at night or in the early morning.

Common Triggers

- Allergens: dust mites, pollen, pet dander, mold
- Respiratory infections (cold, flu)
- Physical exercise (exercise-induced asthma)
- Cold air or changes in weather
- Air pollutants and smoke
- Strong emotions and stress
- Certain medications (aspirin, NSAIDs, beta-blockers)

Symptoms

- Shortness of breath
- Wheezing (whistling sound when breathing)
- Chest tightness or pain
- Coughing (worse at night or early morning)
- Difficulty sleeping due to breathing problems

Treatment

- **Quick-relief (rescue) inhalers:** Short-acting beta-agonists (SABA) like albuterol for immediate symptom relief.
- **Long-term control medications:** Inhaled corticosteroids (ICS) like fluticasone — first-line maintenance therapy.
- **Combination inhalers:** ICS + long-acting beta-agonists (LABA) for moderate-to-severe asthma.
- **Biologics:** Monoclonal antibodies (omalizumab, dupilumab) for severe allergic asthma.
- **Avoid triggers** and have a written asthma action plan.

4. Cardiovascular Diseases

Definition

Cardiovascular diseases (CVD) include disorders of the heart and blood vessels, such as coronary artery disease, heart failure, arrhythmias, and stroke. CVD is the leading cause of death worldwide, responsible for approximately 17.9 million deaths per year.

Major Types

- **Coronary Artery Disease (CAD):** Narrowing of coronary arteries due to atherosclerosis, leading to angina or heart attack.
- **Heart Failure:** The heart cannot pump enough blood to meet the body's needs.
- **Arrhythmias:** Abnormal heart rhythms such as atrial fibrillation.
- **Stroke:** Interruption of blood supply to the brain (ischemic or hemorrhagic).
- **Peripheral Artery Disease:** Reduced blood flow to the limbs due to narrowed arteries.

Risk Factors & Prevention

- Control hypertension and diabetes
- Maintain healthy cholesterol levels (LDL < 100 mg/dL)
- Quit smoking and limit alcohol
- Regular aerobic exercise (150 min/week)
- Heart-healthy diet (Mediterranean or DASH diet)
- Medications: statins, antiplatelets, antihypertensives when indicated

5. Obesity

Definition

Obesity is a complex disease involving excessive body fat that increases the risk of other diseases and health problems, such as heart disease, diabetes, high blood pressure, and certain cancers. It is diagnosed using the Body Mass Index (BMI): a BMI of 30 or higher is classified as obese.

BMI Classification

Category	BMI Range
Underweight	< 18.5
Normal weight	18.5 – 24.9
Overweight	25.0 – 29.9
Obese (Class I)	30.0 – 34.9
Obese (Class II)	35.0 – 39.9
Extremely Obese (Class III)	>= 40.0

Treatment

- Caloric restriction and balanced diet (500–750 kcal deficit/day)
- Regular physical activity (150–300 min/week moderate intensity)

- Behavioral therapy and counseling
- Pharmacotherapy: orlistat, semaglutide (GLP-1 agonists)
- Bariatric surgery for BMI ≥ 40 or ≥ 35 with comorbidities

This document is intended for use as a RAG knowledge base. Always consult a qualified healthcare professional for medical advice.